Aliter method:

We know the identity, $(a+b)^2 = (a+b)(a+b)$

$= a^2 + 2ab + b^2$

	a	b
a	a^2	ab
b	ab	b^2

To find the value of 21^2,

We take, $21^2 = (20+1)^2$

$= (20+1)(20+1)$

Here, $a = 20$ and $b = 1$.

Therefore,

$a^2 + 2ab + b^2 = 20^2 + 2\times 20\times 1 + 1^2$

$= 400 + 40 + 1 = 441$.

3.3.3 Identity-3: $(a-b)^2 = a^2 - 2ab + b^2$

When we replace 'b' by '$-b$' in 'identity-2', we get a new identity.

	a	$-b$
a	a^2	$a\times -b = -ab$
$-b$	$a\times -b = -ab$	$-a\times -b = b^2$

We know that, $(a+b)^2 = a^2 + 2ab + b^2$

Taking 'b' as '$-b$' in 'identity-2', we get, $[a+(-b)]^2 = a^2 + 2a(-b) + (-b)^2$

$(a-b)^2 = a^2 - 2ab + (-b)(-b)$

Therefore, $(a-b)^2 = a^2 - 2ab + b^2$

Note that, when we change the sign of b, the sign of $2ab$ (second term) alone is changed.

Example 3.3 Using the identity $(a-b)^2 = a^2 - 2ab + b^2$, simplify the following:

(i) $(3x-5y)^2$ (ii) 47^2

Solution

(i) $(3x-5y)^2$

Put $a = 3x$ and $b = 5y$ in the identity $(a-b)^2 = a^2 - 2ab + b^2$, we get,

$(3x-5y)^2 = (3x)^2 - 2\times(3x)\times(5y) + (5y)^2$

$= 3^2 \times x^2 - (2\times 3\times 5)xy + (5^2 \times y^2)$

$= 9x^2 - 30xy + 25y^2$.

(ii) $47^2 = (50-3)^2$, substituting $a = 50$ and $b = 3$ in the identity $(a-b)^2 = a^2 - 2ab + b^2$, we get,

$(50-3)^2 = 50^2 - 2\times 50\times 3 + 3^2$

$= 2500 - 300 + 9$

$= 2509 - 300 = 2209$.

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